

# L13: Horvitz-Thompson Estimation of "Node Totals"

Let us now use the previous inclusion probabilities to estimate various "node totals", i.e., **metrics that are defined based on the summation of a metric across all nodes**.

Specifically, suppose that each node  $i$  has some property  $y_i$ . We often want to estimate the total value of that property across the whole network:

$$\tau = \sum_{i \in V} y_i$$

For instance, if  $y_i$  is the degree of each node, then we can use  $\tau$  to calculate the average node degree (just divide  $\tau$  by the number of nodes).

Or, if  $y_i$  represents whether a node of a social network is a bot or not (a binary variable), then  $\tau$  is the total number of bots in the network.

Or, if  $y_i$  represents some notion of node capacity, then  $\tau$  is the total capacity of the network.

Suppose that we have a sample  $S$  of  $n$  network nodes, and we know the value  $y_i$  of each sampled node.

An important result in statistics is the **Horvitz-Thompson** estimator. It states that we can estimate  $\tau$  as follows:

$$\hat{\tau} = \sum_{i \in S} \frac{y_i}{\pi_i}$$

where  $\pi_i$  is the inclusion probability of node- $i$ , as long as  $\pi_i > 0$  for all nodes in the sample  $S$ .

In other words, to estimate the "total value" we should not just add the values of the sampled nodes and multiply the sum by  $N/n$  (that would be okay only if  $\pi_i = \frac{n}{N}$  for all  $i$ ). Instead, *we need to normalize the value of each sampled node by the probability that that node is sampled*.

A key property of the Horvitz-Thompson estimator is that it is *unbiased*, as shown below. Let us define as  $Z_i$  an indicator random variable that is equal to one if node  $i$  is sampled in  $S$  -- and 0 otherwise. Then,

$$E[\hat{\tau}] = E\left[\sum_{i \in S} \frac{y_i}{\pi_i}\right] = E\left[\sum_{i \in V} \frac{y_i Z_i}{\pi_i}\right] = \sum_{i \in V} \frac{y_i E[Z_i]}{\pi_i} = \sum_{i \in V} y_i = \tau$$

An estimate of the variance of this estimator can be calculated from the sample  $S$  as follows:

$$V(\hat{\tau}) \approx \sum_{i \in S} \sum_{j \in S} y_i y_j \left( \frac{\pi_{i,j}}{\pi_i \pi_j} - 1 \right)$$

where  $\pi_{i,j}$  is the probability that both nodes  $i$  and  $j$  are sampled in  $S$  (when  $i=j$ , we define that  $\pi_{i,j}=\pi_i$ ).

## Food For Thought

Suppose that  $i$  and  $j$  are sampled independently. How would you simplify the previous expression for the variance of the Horvitz-Thompson estimator?